

Brainstorming & Idea Prioritization Template

Date	03 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach for Flood Prediction
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Sasi Vardhan Reddy	Build a predictive system using historical weather data (rainfall, cloud visibility) and ML classification models (XGBoost, RF) integrated with a Flask web app.	Machine Learning & Predictive Analytics	Group 1
2	Sasi Vardhan Reddy	Deploy a real-time IoT network using ultrasonic depth sensors along major riverbeds to monitor water level spikes.	IoT & Hardware Infrastructure	Group 2
3	Sasi Vardhan Reddy	Utilize deep learning and Satellite Remote Sensing (SAR) imagery to map and track flooded zones post-disaster.	Computer Vision & Remote Sensing	Group 3
4	Sasi Vardhan Reddy	Develop an automated SMS and Telegram broadcast system that alerts local authorities when rainfall crosses a specific millimeter threshold.	Notification & Cloud Automation	Group 4
5	Sasi Vardhan Reddy	Create a crowdsourced mobile application where citizens can report live street-level waterlogging to assist disaster response teams.	Mobile Application & Crowdsourcing	Group 4

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Step 2: Idea Prioritization

This section justifies why our team chose the Machine Learning approach over the others based on its high feasibility (using open-source data like Kaggle) and high impact.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1.	ML-based flood prediction using historical weather data (XGBoost, KNN, RF, DT) with Flask deployment. IoT-based real-time river water level monitoring. Satellite imagery analysis for post-flood mapping. Crowd-sourced flood alert and reporting application.	High (Data available on Kaggle; uses standard open-source ML libraries like Scikit-Learn).	High (Provides crucial proactive early warning systems for evacuation planning).	1	Yes
2.		Low	High	2	No
3.		Medium	Medium	3	No
4.		Medium	Medium	4	No